Joseph Walton

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Professional Summary

AI and Machine Learning Engineer with over 10 years' experience spanning software engineering, production AI platforms and business-facing delivery. Strong Python background with hands-on experience building LLM-powered agents, RAG applications, evaluation frameworks, observability capabilities and secure AI services using Azure OpenAI, Azure AI Foundry, LangGraph and LangChain. Proven ability to own solutions end to end, translate operational needs into governed production systems and deliver substantial business value. Experienced in financial services, central government, defence and other regulated environments.

Technical Skills

- **AI & Agents:** Azure OpenAI, Azure AI Foundry, LangGraph, LangChain, RAG, tool-using agents, prompt engineering, Hugging Face
- **AI Quality & Security:** LLM evaluation, observability, human-in-the-loop workflows, red teaming, adversarial testing, responsible AI
- **Software Engineering:** Python, SQL, Java, service-oriented architecture, REST services, testing, CI/CD, Git, Docker
- **Data Platforms:** Apache Spark, PySpark, Polars, Pandas, Azure Databricks, Elasticsearch, vector search
- **MLOps:** Azure ML Pipelines, Vertex AI Pipelines, MLflow, automated evaluation and deployment
- **Cloud:** Microsoft Azure, Google Cloud Platform, AWS
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, NLP, recommendation systems, computer vision

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Professional Experience

IBM

Lead Data Scientist
June 2025 - Present | Contract, Central Government

Delivering production AI solutions and improving engineering practices within a regulated government environment.

- Rescued underperforming AI projects by introducing robust evaluation and observability practices and challenging approaches with evidence.
- Designed and delivered an LLM-powered data-processing agent using Python, Azure AI Foundry, Azure Functions and LangGraph.
- Translated caseworker requirements into a production solution saving approximately two hours per user each day and £2M annually.
- Developed a Claude Skill and human-in-the-loop workflow that reduced preparation time for LLM evaluation cases by approximately one hour per case.
- Re-engineered a Spark data-matching process in Polars, reducing runtime from two hours to ten minutes and cloud expenditure by approximately £10,000 per month.

Technology: Python, Azure AI Foundry, Azure Functions, LangGraph, Polars, Apache Spark

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Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Led the development and operationalisation of secure AI and NLP products within a regulated public-sector environment.

- Designed and built generative AI services using Python, Azure OpenAI, LangChain and Azure Functions within a service-oriented architecture.
- Built a RAG agent that used governed organisational data to help users identify relevant trade mark Goods and Services terms.
- Established a product roadmap for AI applications, balancing user value, delivery constraints, security and responsible-AI risks.
- Conducted red-team testing of LLM applications, identifying adversarial and prompt-based failure modes and improving system robustness.
- Developed evaluation approaches for AI scaffolding, search and recommendation features, supporting iterative, evidence-led improvement.
- Implemented and tuned Elasticsearch HNSW vector search, balancing recall, throughput and operational performance.
- Improved search and recommendation relevance metrics by 20%.
- Implemented an end-to-end MLOps strategy covering testing, deployment and promotion across environments, reducing deployment time to under one day.
- Architected Python and Polars ETL services to replace Azure Databricks workloads, reducing compute costs by 50% and enabling zero-downtime deployments.
- Communicated AI strategy and implementation to technical and non-technical stakeholders across international IP offices.
- Contributed to cross-government AI policy and engineering best practices.

Technology: Python, Azure OpenAI, LangChain, Azure Functions, Azure ML Pipelines, Elasticsearch, Polars, Azure Databricks

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Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led production ML engineering for intellectual-property protection products.

- Designed an MLOps platform on Google Cloud using Vertex AI Pipelines to orchestrate training, evaluation, approval and deployment.
- Reduced model time to market to under one day while supporting client-specific deployments.
- Built Spark pipelines for large-scale text and image processing, training, batch inference and recommendation workloads.
- Developed and deployed a multilingual BERT system that detected potential IP infringements across more than 90 languages.
- Built recommendation capabilities combining text and image similarity.
- Mentored a software engineer transitioning into machine learning.

Technology: Python, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, GCP, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Engineered a production NLP system for predicting IP infringement in a secure, containerised environment.
- Increased enforced infringements by 10% per analyst across a workforce of approximately 200 analysts.
- Built and deployed an image-similarity capability that helped secure a major client contract worth 10% of annual recurring revenue.

Technology: Python, TensorFlow, FastText, Docker, GCP, MLflow

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Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Developed scalable Python and PySpark services to link and process sensitive government administrative data.
- Contributed to the highly scalable backend supporting the UK Census 2021 data-collection system.
- Worked within strict security, privacy and data-governance boundaries.

Technology: Python, PySpark, Cloudera

Barclays

Data Scientist

September 2018 - October 2018 | Secondment

- Applied statistical and analytical techniques to regulated payments data to develop regional economic indicators.
- Continued work originating from the winning ONS x Barclays Data Hackathon entry.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

- Developed full-stack and big-data processing systems for government and defence clients.
- Built production software using Java, Python, PySpark and AngularJS in security-sensitive environments.

Education

University of Bath

MSc Applied Mathematics - *2016-2017*

BSc Physics - *2010-2013*

Award

Winner, ONS x Barclays Data Hackathon
Used payments data to create more granular and timely estimates of Gross Value Added.